WORKED SOLUTION

STEP I 2019, Question 7

Pure • Irrationality and Congruences

STEP I 2019, Question 7

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

If $\sqrt{2}+\sqrt{3} = a/b$ in lowest terms, squaring twice clears the surds and leaves a polynomial equation in the integers a and b. Then we look at that equation modulo a small prime. The equation forces the prime to divide both a and b or neither. "Both" contradicts lowest terms, and remainders rule out "neither". The whole art is choosing a prime for which the remainders really do clash.

Solution

Part (i)

Step 1. Every integer is $3k$, $3k+1$ or $3k-1$ for some integer k, so if $3 \nmid a$ then $a = 3k \pm 1$. In either case

$$a^2 = 9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1,$$

which is one more than a multiple of 3.

Step 3. Squaring $\dfrac{a}{b} = \sqrt{2}+\sqrt{3}$ gives

$$\frac{a^2}{b^2} = 2 + 2\sqrt{6} + 3, \qquad \text{so} \qquad a^2 - 5b^2 = 2\sqrt{6}\,b^2.$$

The surd is now on its own, so squaring again removes it:

$$a^4 - 10a^2b^2 + 25b^4 = 24b^4,$$
$$a^4 + b^4 = 10a^2b^2,$$

as required. (We only need the implication in this direction: any a, b from step 2 must satisfy this equation.)

Step 4. Suppose $3 \mid a$. Rearranging step 3,

$$b^4 = 10a^2b^2 - a^4 = a^2\,(10b^2 - a^2),$$

and the right-hand side is divisible by 3 (even by 9). If $3 \nmid b$, then step 1 gives $b^2 = 3m+1$, so $b^4 = (b^2)^2 = 9m^2+6m+1$ is *not* a multiple of 3, a contradiction. Hence $3 \mid b$.

Ask yourself Step 4 handles the case $3 \mid a$. What about the case where 3 divides neither a nor b?

Step 5. The equation in step 3 is symmetric in a and b, so step 4 also works with the roles swapped: if $3 \mid b$ then $3 \mid a$. Since a and b have no common factor greater than one, they are not both multiples of 3, so by step 4 neither of them is. This case is not covered by steps 2–4, and it is where step 1 comes in. Write $a^2 = 3r+1$ and $b^2 = 3s+1$. Then a^4, b^4 and a^2b^2 are all one more than multiples of 3 (a product of two such numbers is again of this form). Working modulo 3 in step 3,

$$a^4 + b^4 \equiv 1+1 = 2, \qquad 10a^2b^2 \equiv 1\cdot 1\cdot 1 = 1 \pmod 3,$$

since $10 \equiv 1$. The two sides of step 3 leave different remainders on division by 3, which is impossible. So no such a and b exist. Also, $\sqrt{2}+\sqrt{3} > 0$, so if it were rational it could be written with a, b positive and coprime. Hence $\sqrt{2}+\sqrt{3}$ is irrational.

Part (ii)

Suppose that $\sqrt{6}+\sqrt{7} = a/b$ with a, b positive integers having no common factor greater than one. As before,

$$\frac{a^2}{b^2} = 13 + 2\sqrt{42} \quad \Longrightarrow \quad (a^2 - 13b^2)^2 = 168\, b^4,$$

which expands to $a^4 - 26a^2b^2 + 169b^4 = 168b^4$, that is,

$$a^4 + b^4 = 26a^2b^2.$$

Squares modulo 5. If $5 \nmid a$ then $a = 5k \pm 1$ or $a = 5k \pm 2$, giving

$$a^2 = 5(5k^2 \pm 2k) + 1 \qquad \text{or} \qquad a^2 = 5(5k^2 \pm 4k) + 4.$$

So $a^2 \equiv \pm 1 \pmod 5$, and therefore $a^4 \equiv 1 \pmod 5$.

Divisibility. If $5 \mid a$, then $b^4 = a^2(26b^2 - a^2)$ is a multiple of 5. But if $5 \nmid b$ then $b^4 \equiv 1 \pmod 5$, so we must have $5 \mid b$. By symmetry, $5 \mid b$ implies $5 \mid a$. Since a and b are coprime, neither is divisible by 5.

Contradiction. Now $a^4 \equiv b^4 \equiv 1$, and $a^2b^2 \equiv (\pm 1)(\pm 1) = \pm 1 \pmod 5$. Since $26 \equiv 1$,

$$a^4 + b^4 \equiv 2, \qquad 26a^2b^2 \equiv \pm 1 \pmod 5.$$

But $2 \not\equiv \pm 1 \pmod 5$, so the equation cannot hold. Hence $\sqrt{6}+\sqrt{7}$ is irrational.

Ask yourself Which step breaks if we try to use 3 here instead of 5?

Why 3 fails. The first half of the argument still works modulo 3: if $3 \mid a$ then $3 \mid b^4$, so $3 \mid b$, and so neither a nor b is a multiple of 3. But then $a^4 \equiv b^4 \equiv a^2b^2 \equiv 1 \pmod 3$, and since $26 \equiv 2 \pmod 3$,

$$a^4 + b^4 \equiv 2 \qquad \text{and} \qquad 26a^2b^2 \equiv 2 \pmod 3.$$

The two sides agree, so there is no contradiction. Modulo 3 the equation $a^4 + b^4 = 26a^2b^2$ is satisfied by *every* pair of non-multiples of 3, so divisibility by 3 cannot rule it out. (In part (i) the coefficient was $10 \equiv 1$, and $2 \neq 1$ gave the clash. Here the coefficient is $26 \equiv 2$, so the clash disappears.)

Checks

- Numerically, $x = \sqrt{2} + \sqrt{3} \approx 3.1463$ gives $x^4 - 10x^2 + 1 \approx 0$ (to about 10^{-14}). Similarly $y = \sqrt{6} + \sqrt{7} \approx 5.0952$ gives $y^4 - 26y^2 + 1 \approx 0$. So the two quartic equations are right.
- Try every pair of remainders: the only solution of $a^4 + b^4 \equiv 10a^2b^2 \pmod 3$ is $a \equiv b \equiv 0$, and the same holds for $a^4 + b^4 \equiv 26a^2b^2 \pmod 5$. But $a^4 + b^4 \equiv 26a^2b^2 \pmod 3$ is also solved by $(1,1)$, $(1,2)$, $(2,1)$ and $(2,2)$. This is exactly why 3 fails in part (ii).
- A computer search found no coprime positive $a, b \leq 400$ satisfying either equation.

Common slips

- Stopping after step 4. Step 4 only shows that a and b can't be multiples of 3. You still need step 1 to deal with the case where neither is.
- Claiming $3 \mid b^4 \Rightarrow 3 \mid b$ without a reason. Step 1 supplies it: if $3 \nmid b$ then b^4 leaves remainder 1.
- In part (ii), forgetting that a square can leave remainder 4 (that is, -1) modulo 5, so a^2b^2 can be $\equiv \pm 1$, not just 1.

Beyond STEP

The number $\sqrt{2} + \sqrt{3}$ is a root of $x^4 - 10x^2 + 1$, and the argument here is really a proof that this polynomial has no rational roots. The rational root theorem gives a quicker route: any rational root must be ± 1, and neither works. Reducing a polynomial modulo a prime to show that it has no roots or factors is a standard tool in first-year algebra (for example, in irreducibility tests). Curiously, $x^4 - 10x^2 + 1$ *factorises* modulo every prime, even though it is irreducible over the rationals, and this is a famous example. Allenby's *Rings, Fields and Groups* develops these ideas, and Davenport's *The Higher Arithmetic* is a lovely introduction to congruences.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.